

Engineering College Library Management System

Complete Info Booklet, Navigation Guide & Syllabus Mapping Matrix

🌟 1. Project Overview (Yeh Project Kya Hai?)

Yeh project ek **Engineering College Library Management System (LMS)** hai jise Python 3.x, SQLite relational database, aur modern Data Science / Data Visualization libraries (Matplotlib, Seaborn, Plotly, Folium, Squarify, Scipy, Statsmodels, Scikit-Learn) ke sath direct academic syllabus requirements ko fulfill karne ke liye design kiya gaya hai.



6 Engineering Departments

Computer Science, AIML, Mechanical, Civil, Electronics, aur E&TC branches ki dedicated collections.



54 Technical Books

Core engineering textbooks (Tanenbaum, Cormen, Pressman, Shigley, Theraja) total copies aur shelf availability ke sath.



90 Students (10 States)

1st year se 4th year ke students jo Maharashtra, Karnataka, Gujarat, Delhi, UP, etc. se enrolled hain.



280+ Realistic Loans

12 mahino ka borrowing data jisme on-time returns, active loans, aur overdue late fines shamil hain.

📁 2. Architecture & File Structure (Kaunsi File Kya Kaam Karti Hai?)

File / Folder Name	Type	Pura Kaam & Role (Hinglish)
main.py	Master Controller	Application ka main gate hai. Terminal me clean interactive menu chalata hai, user se choice input leta hai aur operations run karta hai.
db_setup.py	Database Engine	library.db SQLite database banata hai. Agar pehle se nahi hai to 4 tables banakar Faker se realistic Indian data seed karta hai.
preprocessing.py	Data Processor	Raw SQL tables ko Pandas DataFrames me clean karta hai. Issue duration, overdue days, ₹5/day fine, aur PCA feature matrix calculate karta hai.
visualizations.py	Visualization Suite	Isme 19+ custom visualization functions hain jo Matplotlib, Seaborn, Plotly Express, Folium maps, aur Scikit-Learn se charts banate hain.
library.db	SQLite Database	Actual database file jisme departments , students , books , aur transactions store hote hain.
library_export.xlsx	BI Data Source	Power BI ke liye 4 sheets me exported Excel file.
charts/	Output Folder	High-resolution (300 DPI) static PNG charts save hone ki jagah.
interactive/	Output Folder	Interactive web charts aur HTML maps save hone ki jagah jo browser me open hote hain.

3. Navigation Guide: Kahan Click Karne Par Kahan Jaate Hain?

Jab aap command line par `python main.py` run karte hain, to software ka flow is tarah chalta hai:

1 **Terminal Launch:** `python main.py`

System check karta hai ki `library.db` mojood hai ya nahi. Agar nahi hai, to `db_setup.py` automatically tables bana kar 280+ transactions load kar deta hai.

2 **Main LMS Menu Screen (Options 0 se 13)**

Aapke samne 14 options aate hain. Numbers 1 to 10 daily library activities ke liye hain, 11 aur 12 data visualization ke liye, aur 13 Power BI ke liye hai.

3 **Option 11 Press Karne Par → Visualization Submenu Open Hota Hai**

Terminal me ek naya 20-option dashboard khulta hai. Yahan aap kisi bhi specific chart number (1 se 19) ko select karke chart generate kar sakte hain.

4 **Static PNG vs Interactive Browser Visuals**

Options 1 se 14, 17, aur 18 high-res PNG images generate karke `charts/` folder me save karte hain. Options 15, 16, aur 19 automatic aapke default web browser (Chrome/Edge) me interactive dashboard aur maps kholte hain!

5 **Option 0 Press Karne Par → Safe Return / Exit**

Submenu me `0` dabane par wapas Main Menu par aa jate hain, aur Main Menu me `0` dabane par program safely close ho jata hai.

4. Main Menu ke Sabhi Options ka Word-to-Word Working

Option 1: **View Library Catalog**

`view_books()`

Library me kitni books hain, unke authors, department, publisher state, aur shelf availability (e.g. 4/5 copies) ek clean tabular format me display karta hai.

Option 2: Add New Book

add_book()

Librarian se Title, Author, Department ID (1-6), Copies, aur Publisher State poochta hai aur `books` table me safely nayi row insert kar deta hai.

Option 3: Remove / Minus Book

remove_book()

Book ID lene ke baad check karta hai ki kitni copies loan par hain. Isme 2 actions hain: (1) Unloaned copies minus karna, ya (2) Pure record ko delete karna. Agar active loan chal raha hai to system delete allow nahi karta!

Option 4: Search Books

search_books()

Title, Author ya Department ka koi bhi keyword (jaise "Python", "Tanenbaum", "Mechanical") search karke matching books list karta hai.

Option 5 & 6: View & Add Students

view_students() | add_student()

College ke 90 students ka directory dikhata hai (Year 1-4, Department, Home State) aur naye student ko register karta hai.

Option 7: Issue a Book (14-Day Auto Due Date)

issue_book()

Student ID aur Book ID poochta hai. Shelf availability check karta hai (available > 0). Issue date aaj ki set karta hai aur due date exact 14 din baad ki set karta hai. Shelf copies 1 minus ho jati hain.

Option 8: Return a Book (Auto Fine ₹5/day)

return_book()

Transaction ID enter karne par check karta hai ki book late return hui ya on-time. Formula:

`overdue_days = return_date - due_date` . Late hone par ₹5.00/day ke hisaab se fine calculate karke shelf stock me +1 restore karta hai.

Option 9 & 10: 🚨 View Overdue Loans & Student History

`view_overdue_books()` | `student_history()`

Option 9 current delinquent loans aur unka live fine dikhata hai. Option 10 kisi bhi specific student ka pura historical borrowing audit trail dikhata hai.

Option 11: 📊 Open Data Visualization Dashboard

`visualization_menu()`

20 dedicated chart options ka complete analytics dashboard submenu kholta hai!

Option 12: ⚡ Generate All Visual Analytics (One-Click)

`generate_all_visualizations()`

Bina kisi option ko baar-baar select kiye, sabhi 19 charts ko ek click me generate karke `charts/` aur `interactive/` folders me save kar deta hai.

Option 13: 📄 Export Dataset for Power BI

`export_for_power_bi()`

Database ke charo tables ko `library_export.xlsx` me 4 alag sheets me export karta hai.

5. Visualization Dashboard (Submenu ke Sabhi 20 Options)

Opt #	Chart Naam	Library / Technique	Output File	Kya Explain Karta Hai?
1	Collection by Department	Matplotlib Bar Chart	<code>charts/books_by_department.png</code>	Har engineering branch me total kitni book copies hain.
2	Author Analytics	Matplotlib Horizontal Bar	<code>charts/author_distribution.png</code>	Top 10 authors jinki books sabse zyada borrow hui hain.
3	Cleveland Dot Plot	Matplotlib Dot & Line	<code>charts/dot_books_per_dept.png</code>	High data-ink ratio alternative of bar chart.
4	Loan Status Breakdown	Grouped & Stacked Bar	<code>charts/loan_status_breakdown.png</code>	Returned vs Active vs Overdue loan breakdown per branch.
5	Duration Spread	Hist + Overlaid KDE + Violin	<code>charts/duration_distributions.png</code>	Student kitne din tak book apne paas rakhte hain.
6	Statistical Diagnostics	ECDF + Q-Q Plot + Log Trans	<code>charts/statistical_diagnostics.png</code>	Fine distribution ki normality check aur skewness removal.

Opt #	Chart Naam	Library / Technique	Output File	Kya Explain Karta Hai?
7	Axis Orientation	Ridgeline KDE vs Boxplots	charts/vertical_vs_horizontal.png	Horizontal continuous flow vs Vertical discrete boxplots.
8	Proportions & Treemap	Pie + 100% Stacked + Treemap	charts/proportions_and_treemap.png	Whole-to-part ratio aur nested book title borrow volume.
9	Associations & PCA	Grammar of Graphics + PCA 2D	charts/associations_and_pca.png	Overdue vs Fine regression, correlation matrix, aur 2D PCA clusters.
10	Multivariate Pairplot	Seaborn Pairplot	charts/borrowing_pairplot.png	Sabhi numerical metrics ka pairwise correlation matrix.
11	Paired Slope Chart	Matplotlib Slope Lines	charts/due_vs_return_slope.png	Assigned due date (+14d) vs actual return date offset.
12	Time Series & Trends	Rolling Average + Linear Fit	charts/time_series_and_trends.png	Monthly issue volume trend aur 3-month smoothing.

Opt #	Chart Naam	Library / Technique	Output File	Kya Explain Karta Hai?
13	Advanced Time Series	Dose-Response + Decompose	charts/advanced_time_series.png	Fine buildup curve aur additive seasonal decomposition.
14	Activity Heatmap	Seaborn Density Heatmap	charts/activity_heatmap.png	Department vs Month borrowing density with continuous colormap.
15	Geospatial Choropleth	Plotly Express Interactive	interactive/choropleth_map.html	India ke states ka student demographics choropleth map.
16	Geospatial Layers	Folium Leaflet Multi-layer	interactive/folium_map.html	Student home origins (circles) + Campus reading rooms (pins).
17	Proportional Cartogram	Scatter Symbol Map (WGS84)	charts/cartogram_proportional.png	Bubble size proportional to total state borrow volume.
18	Design Principles Demo	Poor 3D Clutter vs Clean Bar	charts/design_principles_comparison.png	Chartjunk elimination aur Proportional Ink principle demo.

Opt #	Chart Naam	Library / Technique	Output File	Kya Explain Karta Hai?
19	Interactive Dashboard	Plotly HTML Interactive	<code>interactive/interactive_dashboard.html</code>	Hover tooltips, zoom, pan, aur filtering in web browser.
20	Batch Render All	Automated Pipeline	All <code>charts/</code> and <code>interactive/</code>	Sabhi visual reports ko ek second me generate karta hai.

💡 6. Viva / Oral Exam Q&A Preparation (Hinglish)

Q1: Aapke project me Edward Tufte ka Data-Ink Ratio principle kahan use hua hai?

Ans: Hamne Option 3 (Cleveland Dot Plot) aur Option 18 (Design Principles Demo) me iska use kiya hai. Unnecessary 3D depth, heavy colored backgrounds, aur redundant grid lines ko hatakar sirf meaningful data points ko ink di gayi hai.

Q2: PCA (Principal Component Analysis) ka library data me kya role hai?

Ans: Har student 6 departments ki books padhta hai (6 dimensions). PCA in 6 features ko compress karke 2 orthogonal components (PC1: Borrowing Volume, PC2: Department Focus) me convert karta hai jisse hum 2D scatter plot par student profiles ko cluster kar sakte hain.

Q3: Q-Q Plot aur ECDF se hum kya check karte hain?

Ans: Q-Q plot se check hota hai ki library fines ka distribution normal distribution ko follow karta hai ya nahi. ECDF se pata chalta hai ki kitne percent students par zero fine hai aur kitne percent par maximum fine laga hai bina kisi binning bias ke.

Q4: Power BI integration kaise kiya gaya hai?

Ans: System SQLite database se 4 relational tables (transactions as Fact Table; books, students, departments as Dimension Tables) ko `library_export.xlsx` me export karta hai, jise Power BI Desktop me Star Schema model me connect karke DAX measures banaye gaye hain.

Engineering College Library Management System & Data Visualization Platform • Built with Python 3.x, SQLite, Matplotlib, Seaborn, Plotly, Folium & Power BI.